

① Hashing :- It is the process of mapping keys and values into the hash table by using a hash function. It is used to store key-value pair.

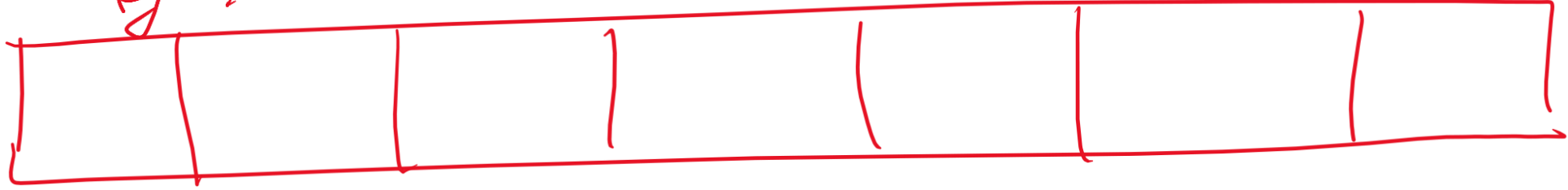
eg :- $\text{map} < \underbrace{\text{String}}_{\text{Key}}, \underbrace{\text{Integer}}_{\text{Value}} > m;$

① $\text{map} [\underbrace{\text{"world"}}_{\text{Key}}] = \underbrace{100}_{\text{value}};$
 $\text{map} [\underbrace{\text{"Cup"}}_{\text{Key}}] = \underline{200};$
 $\text{map} [\underbrace{\text{"IPL"}}_{\text{Key}}] = \underbrace{300}_{\text{value}};$

② The main use of HashMap is efficient retrieval of data;

① How HashMap Works Internally :-

Array :-

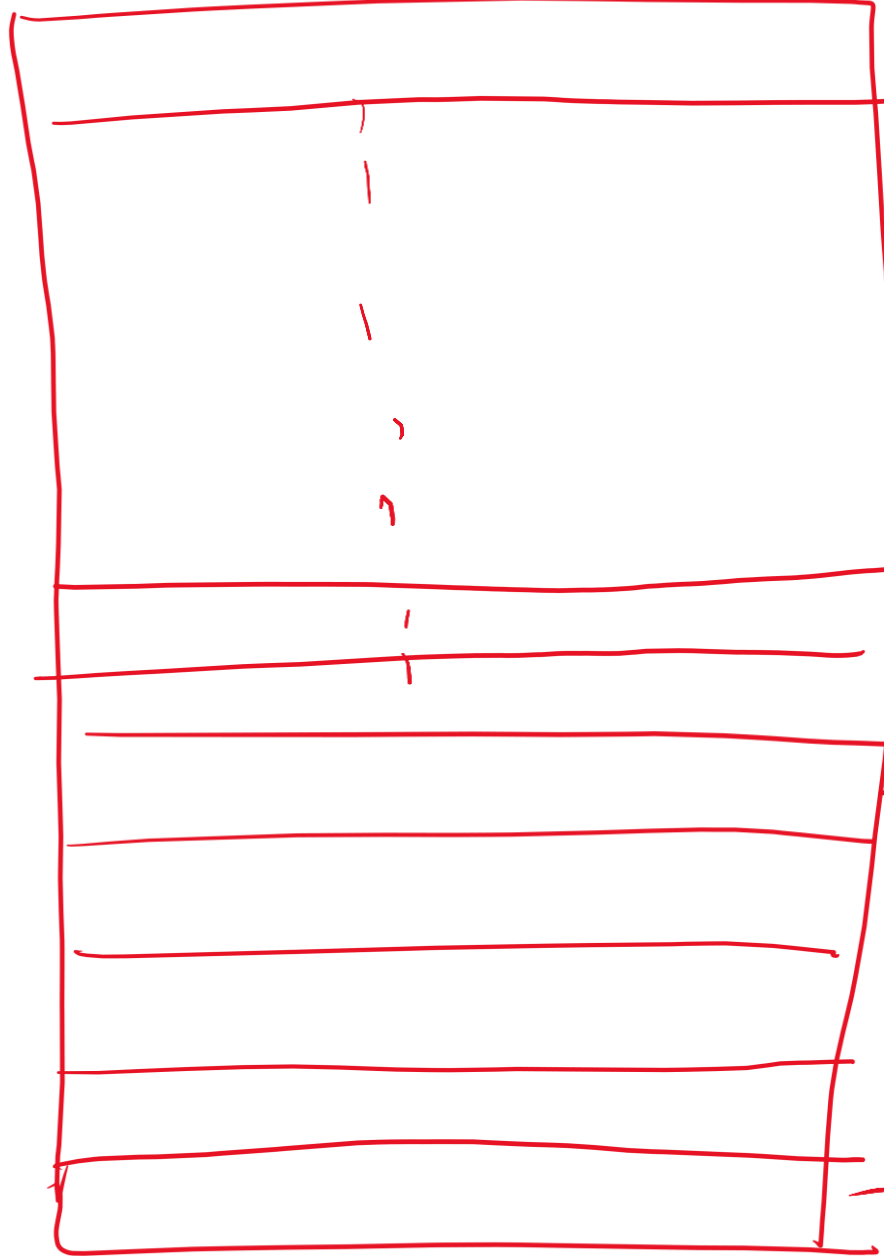


Search T-C = $\theta(n)$

$$n = 10^7 \text{ or } 10^8$$

o HashMap Consists

15 of BUCKETS:-



"CRICKET"

+

6 → CRICKET

4 →

DSA

Hashfunction

70

3 →

S.D

2

1

0

$$70 \% 16 = 6$$

"DSA" →

$$20 \% 16 = 4$$

"S.D" → 4